NR–ReX Network Integration Project

Client: National Railways (NR)

Prepared By Group 03

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Project Plan

Client	National Railways (NR)
Project Name	NR–ReX Network Integration Project
Project Manager	Kavish Gunasekara
Synopsis	This document serves as the final version (Stage 2) of the Project Plan for the NR–ReX Network Integration Project. It outlines the overall project scope, deliverables, approach, communication plan, work plan, milestones, risk management, and financial planning required for successful integration between NR and ReX.
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Distribution	Executive Sponsor, Project Team Members, Project Records
Prepared by	Kavish Gunasekara (Project Manager)
Authorized by	_____*Ochini*_____ (Executive Sponsor) _____*Michael*_____ (NR COO / Business Sponsor)

Amendment History

Version Issue	Issue Date	Changes
Draft V0.1	October 1, 2025	Initial draft created for internal review.
Draft V0.2	October 5, 2025	Incorporated team feedback and updated risk/financial sections.
Final V1.0	October 12, 2025	Approved final version.

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1 Introduction

1.1 Document Purpose

This document presents the Project Management Plan (PMP) for the Network Integration Project between Regional Express (ReX) and National Railways (NR).

The plan defines the resources, overall processes and resources required to integrate ReX's regional operations with NR's national transport network.

It outlines how the project will be planned, executed, controlled and monitored over the 20 month timeframe. The PMP serves as the formal baseline plan for guiding, coordinating stakeholders, project activities and managing progress toward the integration objectives. It is the first approved version of the project plan and will remain the principal reference for scope, communication, scheduling and quality management throughout all phases of delivery.

1.2 Associated Documents

This Project Plan is part of a broader set of documents that collectively define, monitor, and control the NR & ReX Network Integration Project. Each related document supports technical process or a specific management and should be read in conjunction with this plan.

Document Name	Issue Date	Purpose / Description
Project Charter Document	Week 4	Defines project objectives, strategic goals and high level scope established during initiation.
Project Management Plan (this document)	Current Issue	Describes detailed planning, governance structure and control methods for the project.
Quality Plan	Planned for Stage 3	Specifies quality standards, review procedures, and acceptance criteria for all deliverables.
Acceptance Test Plan	To be issued in Stage 3	Outlines acceptance testing methods, verification of final system performance and client sign off process.
Quality and Risk Management Register	To be issued in Stage 2	Identifies project risks, corresponding quality assurance activities and mitigation actions.
Communication Plan	Included in Section 5 of this document	Defines communication channels, reporting frequency and stakeholder responsibilities.
Microsoft Project File	To be issued in Stage 2	Displays the detailed project schedule, milestone tracking and task dependencies.
Project Closure Report	To be prepared in Stage 3	Summarises project outcomes, post implementation maintenance actions and lessons learned

Together with this Project Plan, the above documents control framework and constitute the complete planning for successful delivery of the Network Integration Project.

1.3 Project Plan and Maintenance

The Project Plan will be maintained under formal version control within the project's SharePoint repository to ensure that all the revisions are approved appropriately and documented.

Major modifications, including changes to schedule baselines or cost, scope will require reapproval from the Business Sponsor and the Executive Sponsor before a new version is issued. Minor updates, such as progress comments or schedule adjustments, will be verified by the Project Manager and noted in weekly status reports.

Each update will include a change log recording the version number, , author, issue date and approval signature. The plan will be formally reviewed at key milestones Design Approval (M3), Pilot Evaluation (M5), and Project Closure (M7) to ensure to incorporate any approved changes and continued alignment with project objectives.

2 Project Scope

The Network Integration Project aims to align Regional Express (ReX) and National Railways (NR) operations under a unified framework that improves coordination, efficiency, and service delivery across Australia. This section outlines the client's objectives, success criteria, and potential risks that influence the project direction and priorities.

2.1 Outline of Client's Objectives

2.1.1 Objectives

From the client's perspective, this project represents a strategic initiative to consolidate and modernise its national logistics network. National Railways, having recently acquired Regional Express, seeks to integrate ReX's regional last-mile delivery operations with NR's established national freight and passenger systems.

The project is undertaken to:

- Eliminate duplication in fleet management, tracking systems and scheduling.
- Improve communication between depots, and operations staff and scheduling teams.
- Enhance route efficiency and reduce logistics costs through advanced digital tracking technologies.
- Strengthen overall service delivery, enabling faster transit times and better visibility for customers.

This initiative fits within NR's long-term strategy to build a unified national transport network, optimising operational resources while expanding regional delivery capacity through ReX's expertise. The integration will also support NR's goal of achieving an annual logistics cost reduction of approximately $850,000 and strengthening its reputation as a leader in sustainable transport operations.

2.1.2 Success Criteria

The success of the client's project will be measured through clear, operational and financial outcomes once integration is complete.

Key indicators include:

- Operational Alignment: All ReX depots and scheduling systems successfully transitioned into NR's unified operational framework.
- Efficiency Gains: Noticeable improvement in coordination between national and regional routes, resulting in reduced service delays and better fleet utilisation.
- Technology Integration: Stable deployment of advanced tracking systems, ensuring real time visibility of assets and deliveries.
- Financial Benefits: Achievement of the projected AUD 850,000 annual cost savings through reduced overhead and process standardization.
- Customer Experience: Leading to improved satisfaction ratings, enhanced transparency, and faster delivery times.

2.1.3 Risks

From the client's perspective, several major risks could affect project success:

- System Compatibility: Differences between Nation Railway and ReX's existing systems may cause integration challenges.
- Operational Disruption: Temporary service interruptions may occur testing phases and during migration.
- Accuracy and Data Quality: Incomplete legacy data or Inconsistent could impact performance of new systems.
- Staff Resistance to Change: Some employees may be reluctant to standardized procedures or adopt new digital tools.
- Supply and Vendor Delays: Delays in third party API or software support may slow progress.

This project also acts as a risk mitigation measure for National Railway (NR) by reducing long term operational vulnerabilities and addressing inefficiencies. Through careful planning, phased rollout, and strong communication management, these risks will be actively monitored and minimised.

2.2 Outline of ReX Objectives

2.2.1 Objectives

From the organization's perspective, the National Railway (NR) & ReX Integration Project Team is responsible for delivering the procedural, technical and operational capability that enables National Railways (NR) to achieve its overall business objectives.

The team's key objectives are to:

- Design and implement a unified operational framework that processes with NR's infrastructure and integrates ReX's systems.
- Configure and deploy scheduling platforms and digital tracking that improve coordination across transport hubs and all depots.
- Establish a secure environment and ensure seamless data migration for operational information sharing.
- Conduct staff training sessions and provide training materials to enable NR and ReX employees to adopt new systems effectively.
- Deliver all planned activities within the approved 20 month schedule and in line with quality and compliance requirements.

In the larger context of NR's strategic program, the project team's responsibility is to create the integration capability the systems, resources and procedures that allow National Railway (NR) to realize its cost reduction and wider goals of efficiency improvement. The team will not manage NR's ongoing operations but will ensure that the necessary frameworks and tools are ready for National Railway (NR) to operate independently after project completion.

2.2.2 Success Criteria

The success of the organization's project delivery will be measured by:

- Completion of all deliverables as defined in the Project Charter and this Project Plan.
- Successful client acceptance of the new operational systems and integration testing.
- Demonstrated user readiness through knowledge transfer activities and completion of staff training.
- Delivery within the allocated budget and the approved project schedule.
- Positive evaluation from the National Railway (client) during milestone reviews, particularly at the pilot and final rollout stages.

Achievement of these criteria will confirm that the project team has delivered functional and a complete integration framework that enables National Railway (NR) to achieve its operational and financial objectives.

2.2.3 Risks

At the organization level, several key risks could affect project delivery:

- Technical Integration Challenges: Incompatibility between legacy systems and new integration platforms may require additional configuration or testing time.

- Resource Availability: Delays in vendor support or securing skilled personnel could affect schedule adherence.

- Change Management Gaps: Insufficient end user adoption may limit the effectiveness of implemented systems.

- Budget Constraints: Cost overruns due to unforeseen technical adjustments may impact financial targets.

These risks are being minimised through proactive planning, phased implementation, detailed risk monitoring, and continuous communication with the stakeholders and the client. Each risk will be reviewed at milestone meetings to ensure timely mitigation and tracked in the Risk Register.

2.3 Definitive Scope Statement

The National Railway & ReX Network Integration Project is responsible for the designing, developing, and implementing the operational, technical and organizational capabilities required to unify National Railways (NR) and Regional Express (ReX) into a single, integrated logistics network.

The definitive scope of this project is established in reference to the approved Project Charter Document (Version 1.0) and the Project Management Plan (Version 1.0), which together form the baseline for all the planning, monitoring, and control activities.

Within this framework, the project team is accountable for:

- System Integration and Configuration – Aligning and connecting National Railway and ReX scheduling, tracking systems and fleet management through a unified digital platform.

- Process Standardisation – Developing and documenting new SOPs that harmonies scheduling workflows and depot operations across both organizations.

- Data Validation and Migration – Mapping, cleansing, and securely transferring operational data from ReX legacy systems into National Railway's central environment.

- Change Enablement and Training – Providing user manuals, training sessions and support to ensure staff readiness and adoption of new processes.

- Pilot Implementation and National Rollout – Testing the integrated system at selected depots before phased deployment across every location.

- Performance Reporting and Monitoring – Reporting mechanisms and establishing metrics for evaluating integration success and ongoing performance.

The project will not include the procurement of new trains, or physical infrastructure upgrades, vehicles nor will it address corporate policy or human resources restructuring within National Railway. These elements remain outside the agreed project boundaries.

All scope elements will be managed through controlled change processes. Any request for additional functionality or work beyond the approved baseline will require approval and formal evaluation by the Executive Sponsor prior and the Project Manager to implementation.

3 Deliverables

The deliverables of the National Railway & ReX Network Integration Project define the intangible and tangible outputs that will be formally handed over to National Railways (NR) and those actions or items that National Railway must, in turn, provide to the project team. All due dates are aligned with the approved 20 month project schedule and are subject to confirmation during milestone reviews and detailed planning.

3.1 To Client

The following items represent the formal deliverables that the project team will provide to NR as part of the contracted scope of work.

Deliverable	Description	Target Delivery Date	Status / Type
Project Management Plan (PMP)	Comprehensive plan describing project execution, control, and governance processes.	Month 4 (estimated)	Document
System Integration Design and Architecture	Detailed configuration and interface specifications for unified scheduling and tracking systems.	Month 6 (estimated)	Technical Design
Data Migration and Validation Package	Mapped, cleansed, and verified dataset ready for upload into NR's production environment.	Month 12 (estimated)	Technical Deliverable
Pilot Implementation Report	Outcomes and performance analysis from pilot testing at selected depots.	Month 15 (estimated)	Operational Report
Training and Change Management Materials	Training manuals, user guides, and recorded sessions supporting staff transition.	Month 16 (estimated)	Support Deliverable
Final System Rollout and Handover Pack	Complete documentation of system configuration, disaster-recovery plan, and maintenance procedures.	Month 20	Final Deliverable
Project Closure Report	Summary of outcomes, lessons learned, and formal client sign off.	Month 20	Closure Document

All deliverables will undergo formal review and acceptance by the Executive Sponsor and Business Sponsor according to the Communication Plan and sign-off procedure defined in this Project Plan.

3.2 From Client

The client, National Railways (NR), has specific responsibilities and deliverables essential to the success of the project. These obligations ensure that the project team can proceed on schedule and maintain quality standards.

Client Deliverable / Responsibility	Description / Purpose	Required Timing
Provision of System Access and Legacy Data	Grant controlled access to ReX and NR systems, databases, and relevant historical data for migration and analysis.	Month 2 – Month 5
Review and Approval of Documents	Provide feedback and formal approval on key deliverables (Charter, Design Plan, Pilot Report, Closure Report) within agreed timelines.	Ongoing throughout project
Availability of Subject Matter Experts (SMEs)	Assign depot and operations specialists to assist with requirements clarification, testing, and training validation.	Month 3 – Month 15
Testing Environment and Hardware Support	Supply necessary hardware, network access, and secure environments for integration testing.	Month 6 – Month 12
Single Point of Contact for Project Management Issues	Nominate a central NR representative to coordinate communications and decision-making.	At Project Start
Timely Budget Release and Resource Approvals	Ensure project funding and internal resource allocations are released as scheduled.	As per financial milestones

These client-side deliverables are critical dependencies for the project team's ability to meet scope, quality and schedule commitments. Omissions or Delays or will be addressed and tracked through the project's issue management process.

4 Project Approach

The Network Integration Project will undergo a predictive (Waterfall) methodology/approach.

Within the model, work is done in a series of structured and ordered phases. Every phase has defined/clear objectives and produces explicate outputs that are reviewed/approved before moving on to the next phase.

Throughout the 20-month integration schedule, this structured process ensures effective control, clear accountability/transparency, and smooth progress.

To review deliverables, manage risks, and confirm readiness to proceed; Checkpoint milestones are set at the conclusion of each phase.

4.1 Project Lifecycle Processes

The project will be delivered through 7 main phases, each ending with a formal checkpoint milestone for review/sign-off.

Phase	Main Activities	Checkpoint / Milestone
Initiation (Month 1)	Ensure the confirmation of the approved project charter, ensure that the project team is appointed, confirm identification of stakeholders, and secure authorization to begin.	M1 – Project Initiation Approved
Requirements & Analysis (Month 2 – 4)	Conduct workshops with NR and ReX staff to collect system and business needs, ensure that existing processes are documented, and identify gaps where found.	M2 – Requirements and Analysis Complete
Design & Planning (Month 5 – 6)	Development of system architecture, integration plan, and process designs; Confirmation of the Work Breakdown Structure (WBS) and Gantt chart.	M3 – Design Approved
Build & Data Migration (Month 7 – 12)	Configure systems, clean/migrate data, and perform initial testing in a controlled trial/test environment.	M4 – Build and Migration Completed
Pilot Implementation (Month 13 – 15)	Deployment of integrated systems at limited depots, review/evaluate performance, and collect feedback for refinement and/or improvement where possible.	M5 – Pilot Evaluation Complete
Full Rollout (Month 16 – 19)	Deployment of the final solution across all depots, validate operational stability and finalize and conclude staff training.	M6 – Full Integration Complete
Closure (Month 20)	Final handover, capture lessons learned and obtain official sign off from National Railway executives.	M7 – Project Closure Approved

Requirements Capture and Prototyping:

All functional needs will be gathered through collaborative workshops, interviews, and document analysis/review.

Simple prototypes/workflow diagrams will be utilized during the Design Phase to confirm and communicate requirements with users before development starts.

Integration and Testing Activities:

Testing will follow a staged process unit testing, system testing, and User Acceptance Testing (UAT).
Results from each testing stage will be communicated and reviewed with the client before proceeding to the next phase.

4.2 Project Management Processes

The project will be managed through defined control processes to ensure consistency, quality, and transparency.

- Change Control:
  All requested changes to scope/schedule/cost will be logged in through a change request register.
  Each requested change will be analyzed/reviewed for its impact and requires approval by the Project Manager and Executive Sponsor before implementation of the change can start..

- Schedule/Progress Tracking:
  Tasks/milestones will be tracked and documented in Microsoft Project (Gantt chart).
  Weekly status updates will compare actual progress with baseline plans and identify any delays or variances.

- Cost/Resource Management:
  The Project Manager will observe expenditure against the cost baseline and make sure that resources are allocated effectively throughout the project.

- Risk Management:
  Risks will be documented and recorded in the risk register, reviewed/analyzed during milestone meetings, and managed through contingency planning and mitigation.

- Quality Assurance:
  Every deliverable undergoes peer review and client review to ensure it meets required quality standards before acceptance.

- Performance Reporting:
  Progress summaries, key metrics, and risks update will be presented in weekly team meetings and monthly Steering Committee reports.

The processes here make sure that the project stays on schedule, stays aligned with stakeholder expectations and stays within budget.

4.3 Project Support Processes

Support processes operate across all phases to maintain consistency, reliability, and documentation control.

- Configuration and Document Control:
  All project documents will be stored in the central SharePoint repository with version control, authorship details, and approval history.

- Release and Deployment Management:
  Each build or software release will first be deployed to a test environment, verified, and then promoted to production after approval.

- Issue and Defect Tracking:
  Issues will be logged in a tracking tool (Excel or Jira) with owners, assigned priorities and due dates to ensure timely resolution.

- Knowledge and Training Support:
  Lessons learned, Frequently Ask Questions (FAQs), and training guides will be shared across National Railway (NR) and ReX teams to promote knowledge transfer and user adoption.

- Backup and Recovery:
  All critical data and configuration files will be backed up weekly.
  Recovery procedures will be tested during the Build Phase to confirm reliability.

These ongoing support activities provide structure and stability throughout the project lifecycle and help maintain quality and accountability.

4.4 Organization

The Network Integration Project will function within a defined project organization structure that makes sure that there are clear accountability/transparency, efficient communication, and effective decision-making across both the project team and the client's management.

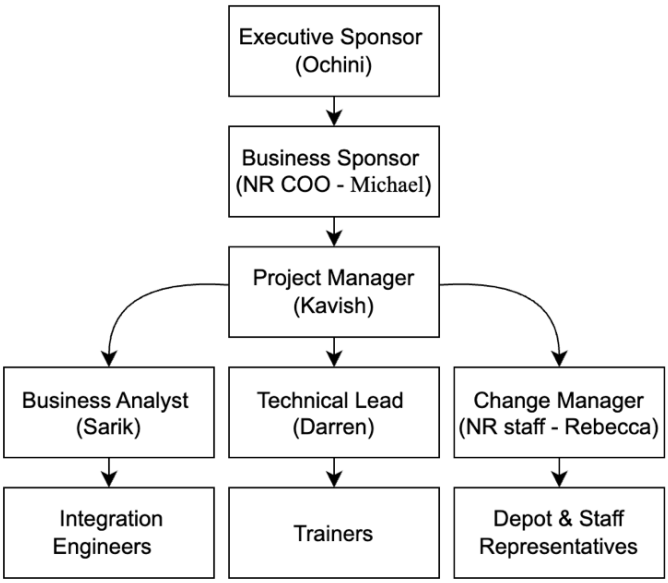
The project will use a predictive (Waterfall) methodology/approach supported by structured review checkpoints and clear role responsibilities.

4.4.1 Project Team

The project team is organized in a hierarchical structure with defined lines of authority and communication.

The Project Manager (Kavish) coordinates all planning, execution, and reporting activities, supported by functional roles including the Business Analyst (Sarik), Technical Lead (Darren), and Change Manager (Rebecca).

Each role has specific responsibilities that contribute to the completion of deliverables and achievement of project objectives.



Role Descriptions

Role / Position	Key Responsibilities
Executive Sponsor (Ochini)	Provides academic oversight, approves major milestones, and ensures adherence to project standards.
Business Sponsor (NR COO - Michael)	Represents client interests, approves scope and budget, and ensures alignment with NR's strategic goals.
Project Manager (Kavish)	Leads project planning, scheduling, risk management, and communication. Ensures deliverables meet objectives.
Business Analyst (Sarik)	Captures requirements, documents business processes, and aligns client needs with technical design.
Technical Lead (Darren)	Oversees system design, integration, and testing activities. Manages data migration and performance validation.
Change Manager (NR Staff - Rebecca)	Coordinates staff training, change communication, and adoption activities.
Integration Engineers / Trainers / Depot Representatives	Provide technical input, conduct testing, support deployment, and assist with training and pilot rollout.

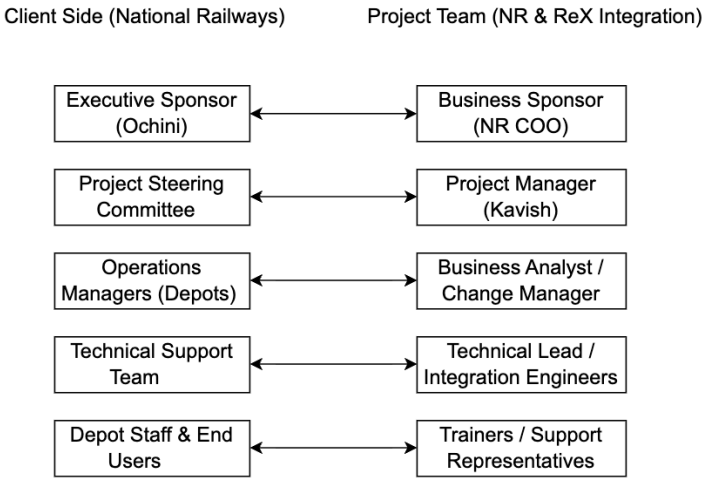
This structure promotes efficient communication, accountability, and cooperation among all parties, ensuring each role contributes to successful project delivery.

4.4.2 Mapping Between ReX and Client

To maintain strong collaboration and oversight, the project follows a multi-level mapping between the project team and the client (National Railways).

This mapping establishes clear reporting and communication links at the executive, management, and operational levels.

Organisational Mapping Diagram



Mapping Explanation

Level	Client Representative	Project Team Contact	Purpose of Interaction
Executive Level	Executive Sponsor (Lecturer) & Business Sponsor (NR COO)	Project Manager (Kavish)	Approve scope, milestones, and overall project direction.
Management Level	NR Steering Committee & Operations Managers	Project Manager & Business Analyst	Review progress, manage risks, and ensure operational alignment.
Technical Level	NR Technical Support & IT Teams	Technical Lead & Integration Engineers	Coordinate system integration, data migration, and testing.
Operational Level	Depot Supervisors & Staff	Trainers & Change Manager	Deliver training, gather feedback, and support day to day adoption.

This mapping structure ensures that communication flows efficiently across all levels, issues are escalated appropriately, and decision-making remains transparent.

Regular meetings between corresponding levels will take place weekly or at milestone reviews, depending on project phase.

5 Communications Plan

Effective communication is essential for the success of the NR & ReX Network Integration Project.

This plan defines who needs information, what information they need, why they need it, when they receive it, and how it will be provided.

It ensures that all stakeholders are informed, decisions are documented, and progress is transparent throughout the 20-month project.

Communication Matrix

Stakeholder	What Information They Need	Why They Need It	When They Receive It	How They Receive It
Executive Sponsor (Lecturer)	Project progress summary, milestone approvals, risk updates	To monitor project performance and approve key deliverables	At major milestones (M1, M3, M5, M7) and monthly summaries	Formal progress reports via email and review meetings
Business Sponsor (NR COO)	Financial performance, integration status, ROI updates	To ensure project outcomes align with NR's strategic and budget objectives	Quarterly reviews and milestone meetings	Executive briefing reports and Steering Committee meetings
Project Manager (Kavish)	Task status, issues, risks, change requests	To manage schedule, resources, and scope control	Weekly team meetings and ad hoc updates	Microsoft Teams, SharePoint reports, and emails
Business Analyst (Sarik)	Requirement updates and change requests	To keep documentation accurate and reflect evolving process needs	During requirement reviews and design phases	Shared documents on SharePoint and review workshops
Technical Lead (Darren)	Technical issues, testing results, build progress	To manage integration activities and technical risks	Weekly technical stand-ups and testing phases	Jira issue tracker and Teams calls
Change Manager (NR Staff)	Training schedules, staff readiness, feedback	To prepare teams for adoption and track change impact	During pilot and rollout phases	Email updates, training sessions, staff surveys
Steering Committee (Mixed NR & ReX Leads)	Summary of progress, budget, risks, decisions required	To provide oversight and approve critical decisions	Monthly Steering Committee meetings	Formal presentation slides and meeting minutes

Depot Managers / Staff	Operational changes, training plans, new procedures	To understand and apply the integrated system effectively	Prior to pilot and full rollout	Workshops, training materials, and email notifications
Vendors / Technical Partners	API updates, support tickets, release notes	To coordinate integration support and resolve issues	As needed during integration and testing phases	Vendor portal and scheduled support calls

Communication Process and Effectiveness

- Frequency:
  Team meetings held weekly; Steering Committee reviews monthly; Sponsor approvals at milestone checkpoints.

- Channels:
  Primary communication tools include Microsoft Teams (for meetings and messaging), SharePoint (for document storage and version control), and email (for formal notifications and approvals).

- Monitoring and Feedback:
  The Project Manager monitors communication effectiveness through attendance records, response times, and feedback from stakeholders. Any communication gaps will be discussed in weekly meetings and corrected promptly.

- Escalation Process:
  Unresolved issues or delays are first escalated to the Project Manager and, if necessary, to the Business Sponsor or Executive Sponsor for review and decision.

This structured communication approach ensures that every stakeholder remains informed, engaged, and aligned with the project's objectives at all times.

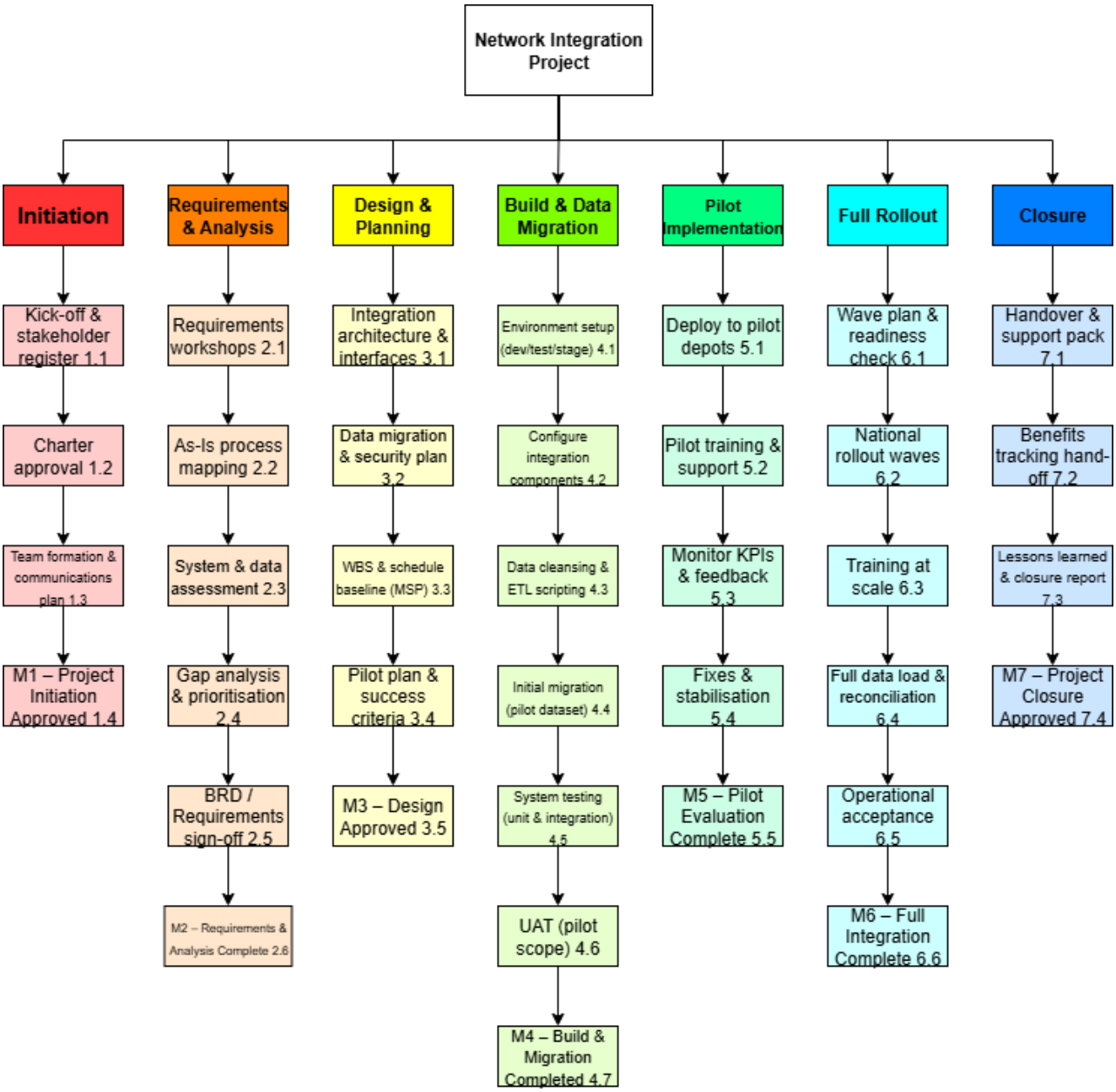
6 Work Plan

This section outlines the tasks required to complete the NR & ReX Network Integration Project and identifies how project resources will be used to deliver them.

The plan is divided into seven main stages, each ending with a milestone to mark key achievements.
The detailed timeline and task dependencies will be displayed in the Gantt Chart created using Microsoft Project.

6.1 Work Breakdown Structure (WBS)

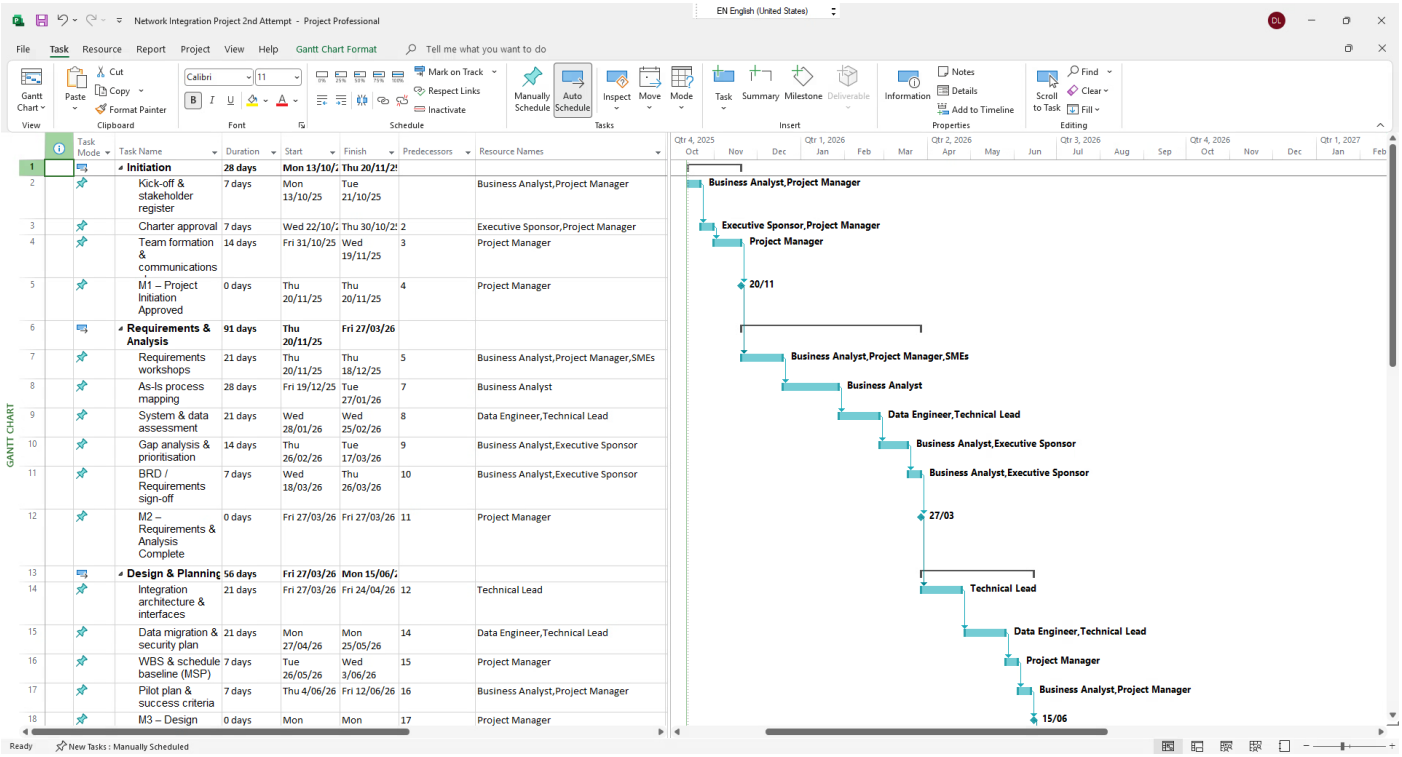
The WBS divides the project into manageable stages and tasks that can be scheduled, monitored, and controlled.



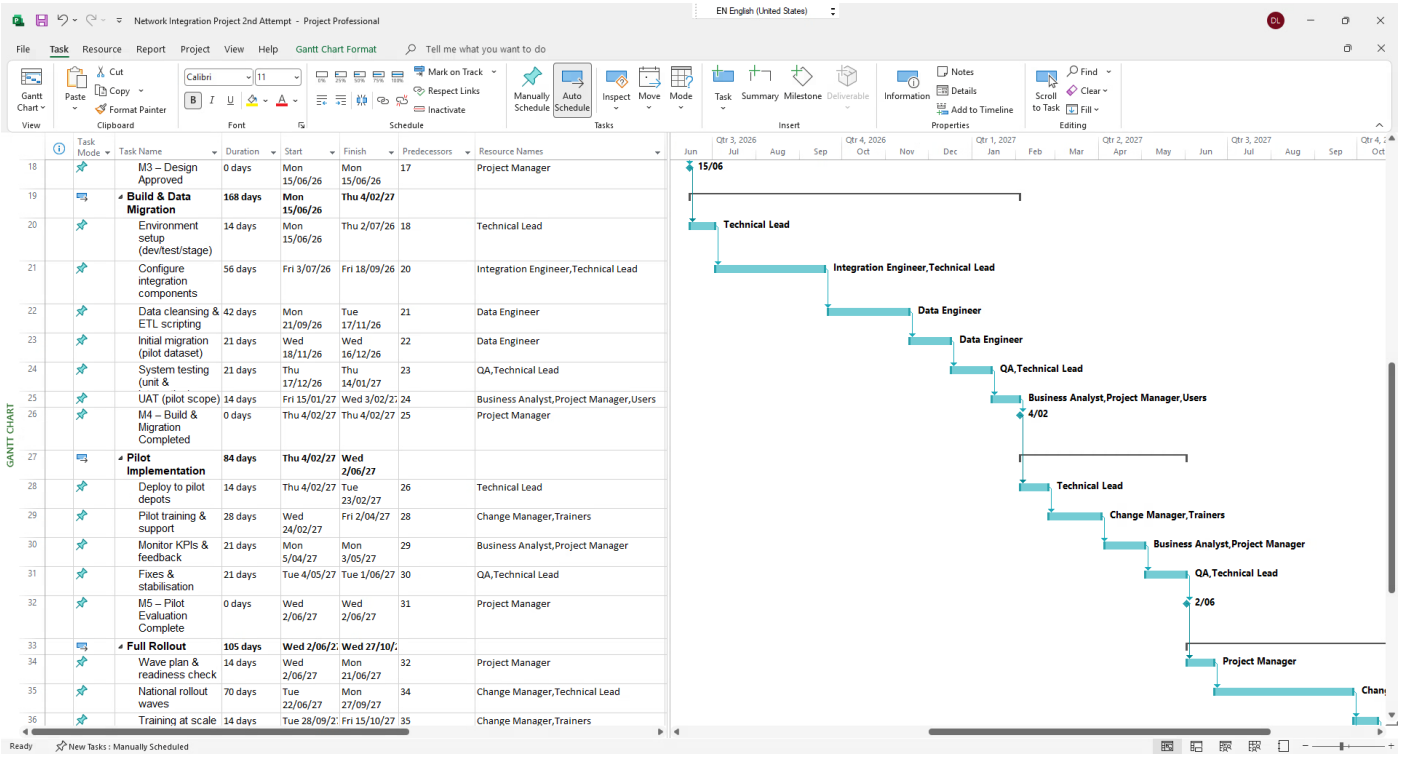
Ongoing activities across all stages:

- Regular progress reporting, risk and issue management, document control, and quality reviews.
- These will continue through every stage but are not separate deliverables.

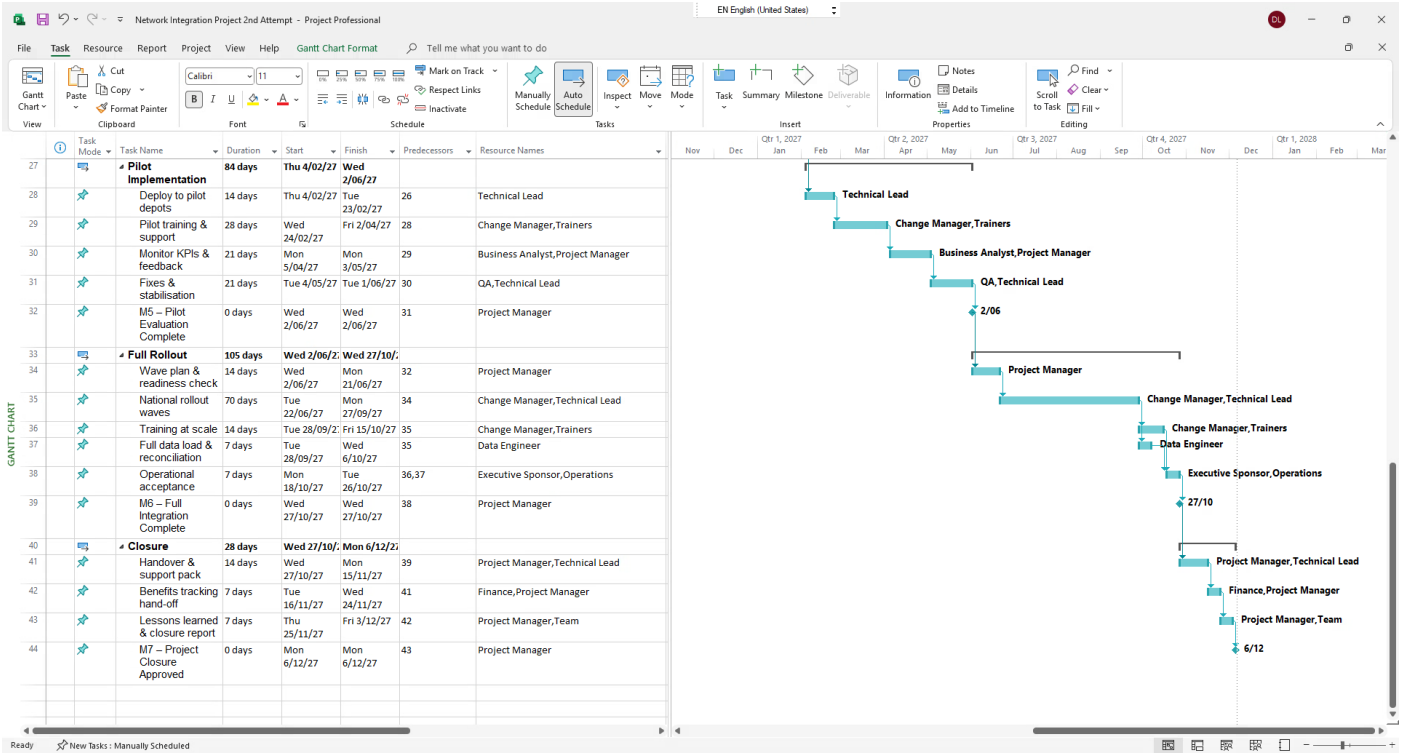
6.2 Resources



	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**Initiation**	**28 days**	**Mon 13/10/**	**Thu 20/11/2**		
2	Kick-off & stakeholder register	7 days	Mon 13/10/25	Tue 21/10/25		Business Analyst,Project Manager
3	Charter approval	7 days	Wed 22/10/	Thu 30/10/2	2	Executive Sponsor,Project Manager
4	Team formation & communications	14 days	Fri 31/10/25	Wed 19/11/25	3	Project Manager
5	M1 – Project Initiation Approved	0 days	Thu 20/11/25	Thu 20/11/25	4	Project Manager
6	**Requirements & Analysis**	**91 days**	**Thu 20/11/25**	**Fri 27/03/26**		
7	Requirements workshops	21 days	Thu 20/11/25	Thu 18/12/25	5	Business Analyst,Project Manager,SMEs
8	As-Is process mapping	28 days	Fri 19/12/25	Tue 27/01/26	7	Business Analyst
9	System & data assessment	21 days	Wed 28/01/26	Wed 25/02/26	8	Data Engineer,Technical Lead
10	Gap analysis & prioritisation	14 days	Thu 26/02/26	Tue 17/03/26	9	Business Analyst,Executive Sponsor
11	BRD / Requirements sign-off	7 days	Wed 18/03/26	Thu 26/03/26	10	Business Analyst,Executive Sponsor
12	M2 – Requirements & Analysis Complete	0 days	Fri 27/03/26	Fri 27/03/26	11	Project Manager
13	**Design & Planning**	**56 days**	**Fri 27/03/26**	**Mon 15/06/**		
14	Integration architecture & interfaces	21 days	Fri 27/03/26	Fri 24/04/26	12	Technical Lead
15	Data migration & security plan	21 days	Mon 27/04/26	Mon 25/05/26	14	Data Engineer,Technical Lead
16	WBS & schedule baseline (MSP)	7 days	Tue 26/05/26	Wed 3/06/26	15	Project Manager
17	Pilot plan & success criteria	7 days	Thu 4/06/26	Fri 12/06/26	16	Business Analyst,Project Manager
18	M3 – Design	0 days	Mon	Mon	17	Project Manager



	Task Name	Duration	Start	Finish	Predecessors	Resource Names
18	M3 – Design Approved	0 days	Mon 15/06/26	Mon 15/06/26	17	Project Manager
19	**Build & Data Migration**	**168 days**	**Mon 15/06/26**	**Thu 4/02/27**		
20	Environment setup (dev/test/stage)	14 days	Mon 15/06/26	Thu 2/07/26	18	Technical Lead
21	Configure integration components	56 days	Fri 3/07/26	Fri 18/09/26	20	Integration Engineer,Technical Lead
22	Data cleansing & ETL scripting	42 days	Mon 21/09/26	Tue 17/11/26	21	Data Engineer
23	Initial migration (pilot dataset)	21 days	Wed 18/11/26	Wed 16/12/26	22	Data Engineer
24	System testing (unit &	21 days	Thu 17/12/26	Thu 14/01/27	23	QA,Technical Lead
25	UAT (pilot scope)	14 days	Fri 15/01/27	Wed 3/02/2	24	Business Analyst,Project Manager,Users
26	M4 – Build & Migration Completed	0 days	Thu 4/02/27	Thu 4/02/27	25	Project Manager
27	**Pilot Implementation**	**84 days**	**Thu 4/02/27**	**Wed 2/06/27**		
28	Deploy to pilot depots	14 days	Thu 4/02/27	Tue 23/02/27	26	Technical Lead
29	Pilot training & support	28 days	Wed 24/02/27	Fri 2/04/27	28	Change Manager,Trainers
30	Monitor KPIs & feedback	21 days	Mon 5/04/27	Mon 3/05/27	29	Business Analyst,Project Manager
31	Fixes & stabilisation	21 days	Tue 4/05/27	Tue 1/06/27	30	QA,Technical Lead
32	M5 – Pilot Evaluation Complete	0 days	Wed 2/06/27	Wed 2/06/27	31	Project Manager
33	**Full Rollout**	**105 days**	**Wed 2/06/2**	**Wed 27/10/**		
34	Wave plan & readiness check	14 days	Wed 2/06/27	Mon 21/06/27	32	Project Manager
35	National rollout waves	70 days	Tue 22/06/27	Mon 27/09/27	34	Change Manager,Technical Lead
36	Training at scale	14 days	Tue 28/09/2	Fri 15/10/27	35	Change Manager,Trainers



Human Resources

The project will rely on both the internal student team and client representatives.

Role	Main Responsibilities / Task Focus
Project Manager (Kavish)	Oversees all phases; manages schedule, budget, communication, and risk register.
Business Analyst (Sarik)	Leads requirements gathering and documentation; supports testing and training.
Technical Lead (Darren)	Designs system architecture, handles integration and data migration tasks.
Change Manager (NR Staff)	Coordinates training, communication, and staff adoption activities.
Integration Engineers / Data Specialists	Execute data cleansing and migration; perform testing and validation.
Trainers / Support Staff	Deliver staff training sessions and provide on-site support during pilot and rollout.

Time allocation will be reflected in the Microsoft Project schedule and adjusted as the project progresses.

Non People Resources

Category	Examples / Purpose	Timeline of Use
Software Tools	Microsoft Project (for scheduling), SharePoint (for document storage), Jira (for issues), MS Teams (for communication)	Throughout project
Hardware / Infrastructure	Laptops, secure servers, test environments, and VPN access for remote collaboration	Mainly Months 5 – 12
Facilities / Training	Meeting rooms, online training platforms, and workshops for pilot and rollout stages	Months 13 – 19
Operational Expenses	Travel for site visits, printing documents, contingency funds for procurement of tools if needed	As required by phase

7 Milestones

Project milestones are the key events that mark the successful completion of major project stages.

They demonstrate progress, approval, and readiness to proceed to the next phase.

Each milestone corresponds to a key deliverable reviewed and approved by the Executive Sponsor and Business Sponsor.

These milestones will also be reflected in the project's Gantt Chart developed using Microsoft Project.

Project Milestone Table

Milestone Number	Milestone Title	Description	Target Completion Date
M1	Project Initiation Approved	Project Charter signed off, stakeholders identified, and team formed.	Month 1
M2	Requirements and Analysis Complete	Requirements gathered and documented, system and data analysis finished, and Business Requirements Document (BRD) approved.	Month 4
M3	Design Approved	Integration architecture, data migration plan, and pilot criteria finalised and approved by stakeholders.	Month 6
M4	Build and Migration Completed	Integration components built, testing completed, and pilot dataset successfully migrated.	Month 12
M5	Pilot Evaluation Complete	Pilot rollout finished, user feedback collected, and readiness confirmed for full deployment.	Month 15
M6	Full Integration Complete	National rollout completed, system operational, and all depots migrated successfully.	Month 19

M7	Project Closure Approved	Handover completed, lessons learned documented, and final approval signed off by sponsors.	Month 20

Notes

- Each milestone represents a formal checkpoint requiring review and sign off by the Executive Sponsor and Business Sponsor.

- The Gantt Chart (developed in Microsoft Project) will include these milestones as zero duration tasks to show when each phase is completed.

- Milestone achievement will be tracked in monthly progress reports and discussed during Steering Committee meetings.

8 Performance and Metrics

Performance measurement ensures that the NR & ReX Network Integration Project achieves its intended outcomes within the approved time, cost, and quality standards.

The project will monitor progress and success through measurable performance indicators linked to key project objectives such as system integration, operational efficiency, user adoption, and financial savings.

8.1 Performance Requirements

The project's main performance requirements are:

- Integration Efficiency:
  Seamless connection between NR and ReX systems with minimal downtime and successful data migration.

- System Reliability:
  The new integrated tracking and scheduling system must maintain a minimum 99.5% uptime during operations and post-rollout.

- Delivery Timeliness:
  All seven project milestones (M1–M7) must be completed within the 20-month timeframe as outlined in the Gantt schedule.

- Cost Control:
  The total project cost must stay within the approved budget, with any variance not exceeding ±10% of planned expenditure.

- Training and Adoption:
  At least 95% of staff should complete training and demonstrate competency in the new digital systems.

- Service Improvement:
  Achieve at least a 10–12% reduction in average route delivery times within six months after the full rollout.

- Customer Satisfaction:
  Reach an 80% or higher satisfaction rating from customers and depot staff in post-implementation surveys.

8.2 Key Metrics and Measurement Approach

Metric	Measurement Method	Collection Frequency	Responsibility	Usage / Purpose
Schedule Performance (Time)	Compare planned vs. actual milestone dates (M1–M7)	Monthly	Project Manager	Identify delays and re-align schedule
Cost Performance	Monitor actual expenditure vs. cost baseline	Monthly	PM & Finance	Ensure project remains within budget
System Uptime (%)	Track system availability using monitoring tools	Continuous (weekly summary)	Technical Lead	Measure stability and performance of integrated platform
Training Completion Rate (%)	Training attendance logs and test results	After each training phase	Change Manager	Ensure readiness and adoption
Efficiency Improvement (%)	Compare average delivery times before and after rollout	Quarterly (post-implementation)	Business Analyst	Measure improvement in logistics efficiency
User / Customer Satisfaction (%)	Feedback surveys and evaluation forms	Pilot phase and final rollout	Change Manager	Assess usability and satisfaction with new systems
Risk Closure Rate	Track number of resolved vs. open risks/issues	Bi-weekly	Project Manager	Evaluate effectiveness of risk management
Data Migration Accuracy (%)	Verify number of errors or data mismatches	During pilot and full rollout	Data Engineers	Ensure data integrity and quality

8.3 Data Collection and Reporting

- Data Sources:
  Project reports, system logs, training records, user feedback surveys, and financial tracking sheets.

- Reporting Frequency:
  Performance reports will be submitted monthly to the Project Steering Committee and reviewed at milestone checkpoints.

- Analysis and Review:
  The Project Manager and Business Analyst will evaluate trends, identify risks, and recommend corrective actions.

- Tools Used:
  Microsoft Project (schedule tracking), Excel/Power BI (performance dashboards), and SharePoint (report storage).

8.4 Use of Metrics

Performance metrics will guide decision making throughout the project.

They will be used to:

- Track progress against baseline objectives.
- Identify early warning signs of schedule or cost overruns.
- Evaluate system performance and adoption rates post-rollout.
- Support lessons learned and continuous improvement during closure.

9 Risks, Constraints, and Assumptions

9.1 Risks

The following table identifies the key risks that may affect the NR & ReX Network Integration Project, along with their contingency plans and mitigation.

These risks will be monitored continuously through the project risk register maintained by the Project Manager (PM) and reviewed at monthly Steering Committee meetings.

Risk ID	Risk Description	Mitigation Plan	Contingency Plan	Impact on Project	Likelihood
R1	System Incompatibility: National Railway and ReX software platforms may not fully integrate, causing data or process conflicts.	Compatibility testing during design and conduct early system audit.	Engage technical consultants or modify APIs to ensure smooth integration.	High: delays and cost overruns.	Medium
R2	Data Quality Issues: missing data or inconsistent in ReX legacy systems may	Perform data cleansing and validation before migration.	Delay cutover, manual corrections or use backup datasets.	High: inaccurate reporting or service issues.	High

	cause migration errors.				
R3	Change Resistance from Staff: employees may struggle to adapt to new digital systems.	Provide workshops, early communication, and incentives for adoption.	Offer additional support sessions and extend training period	Medium: Lower productivity and slower adoption and.	High
R4	Vendor / API Delays: technology vendors may delay updates or API access required for integration.	Maintain strong service level agreements (SLAs) and weekly vendor meetings.	Identify alternate suppliers or temporary manual workarounds.	Medium: milestone delays or partial functionality.	Medium
R5	Budget Overrun: underestimation of costs or unexpected expenses during integration.	Track weekly budget reports and establish 10% contingency reserve.	Reprioritise scope or request phased funding approval.	High: financial impact and stakeholder dissatisfaction.	Low–Medium
R6	Cybersecurity Threats: data breaches or system attacks during migration.	Implement penetration testing and follow ISO 27001 standards.	Activate incident response plan and isolate affected systems.	High: loss of data or trust.	Low
R7	External Disruptions: Natural disasters, industrial strikes or supply chain interruptions.	Monitor external environment and maintain flexible scheduling.	Rebaseline project timeline and activate alternate work arrangements.	Medium: potential delay of activities.	Low

9.2 Constraints

The NR & ReX Network Integration Project operates under the following constraints:

1. Budget Limitations:
   The project must achieve integration and process alignment within a fixed financial limit, with only a 10% contingency buffer allowed.

2. Timeline Constraint:
   All deliverables must be completed within the 20-month period as approved by the Business Sponsor.

3. Tool and Technology Limitations:
   The project must use the existing NR digital platforms, APIs, and infrastructure—no procurement of major

new systems.

4. Data Access Restrictions:
   Access to ReX's legacy data depends on regulatory compliance and privacy requirements, which may restrict full data visibility.

5. Team Availability:
   The project depends on part-time contributions from internal staff, which may affect scheduling flexibility.

9.3 Assumptions

The project plan is based on the following assumptions:

1. Budget Approval:
   The total project budget and contingency funds will be approved and released on schedule.

2. Resource Availability:
   Key personnel (PM, BA, Technical Lead, Engineers, Trainers) will remain available throughout all phases.

3. Stakeholder Support:
   All NR and ReX managers and staff will cooperate and participate actively during integration and training.

4. Stable Regulations:
   No major changes will occur in government transport or digital data compliance laws during the 20-month period.

5. Vendor Cooperation:
   Third-party technology vendors will provide required APIs, documentation, and technical assistance on time.

6. No Major External Disruptions:
   Events such as strikes or natural disasters will not significantly interrupt project execution.

10 Financial Plan

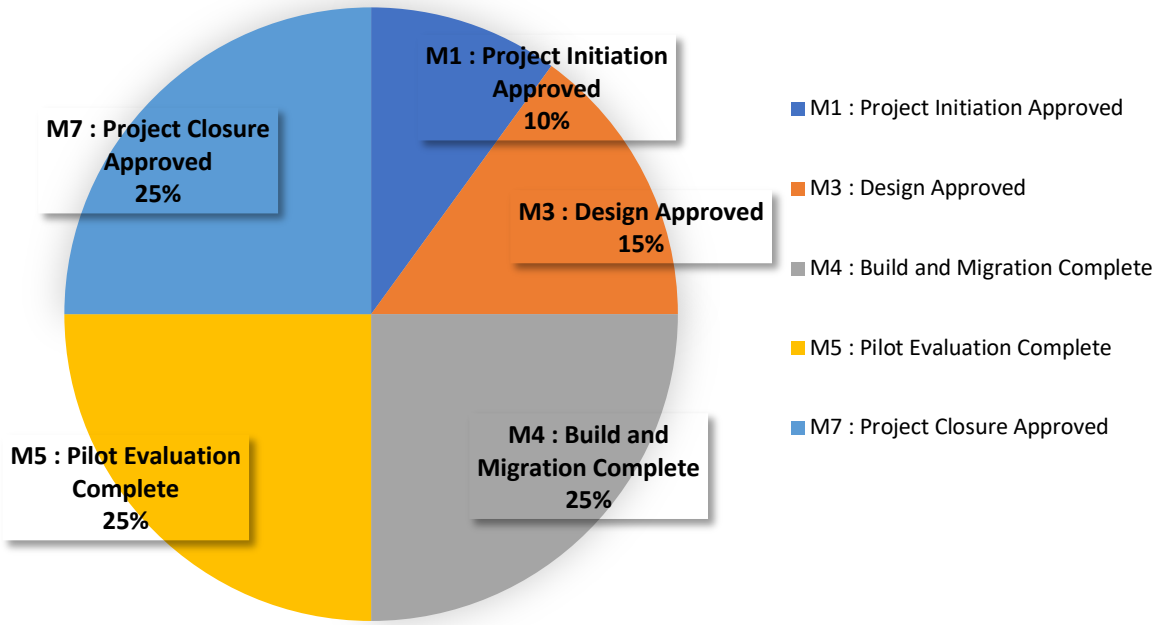
The Financial Plan outlines how the NR & ReX Network Integration Project will be funded, controlled, and reported throughout its 20 month duration.

It ensures that all deliverables are achieved within the approved AUD 850 000 budget while maintaining transparency, efficiency, and accountability.

10.1 Detailed Budget Distribution

NR & ReX Network Integration Project		
Project Budget Distribution		Budget Summary
		Budget (AUD)
01/01/2025 – 31/08/2026 (20 months)		$850,000.00
Tasks	**Unit/ Description**	**Estimated Cost**
Personnel & Labour		$515,000.00
Project Manager	20 months	$120,000.00
Business Analyst	18 months	$100,000.00
Technical Lead	18 months	$110,000.00
Data Engineers	2 persons *12 months	$120,000.00
Trainers / Change Manager	Training design & delivery (pilot + rollout)	$35,000.00
Quality Assurance & Testing Support	Contract / part-time	$20,000.00
Administrative / Support Staff	Shared role	$10,000.00
Software, Tools & Licensing		$70,000.00
Microsoft Project (planning & scheduling)	3 licences	$4,500.00
Jira & SharePoint (issue tracking & collaboration)	Annual subscription	$6,500.00
Data Migration & Integration Tools (API, ETL)	Package licence	$18,000.00
Security & Monitoring Tools (pen-test, logs)	Software package	$10,000.00
Backup & Cloud Storage	Annual plan	$9,000.00
QA & UAT Testing Tools	Pilot & rollout phase	$7,000.00
Licence Renewal / Maintenance Fees	Annual contract	$5,000.00
Hardware & Infrastructure		$65,000.00
Server Hosting & Configuration	2 × medium servers	$15,000.00
Network Equipment (routers, firewalls, VPN)	Implementation cost	$9,000.00
Workstations for Integration & Testing	5 × computers	$11,000.00
Backup Devices & External Storage	3 × devices	$7,000.00
IT Security Hardware (encryption modules)	Set of devices	$8,000.00
Hardware Maintenance & Upgrades	Ongoing	$10,000.00
Training & Change Management		$55,000.00
Training Workshops & Manuals	8–10 sessions	$18,000.00
Online Learning & Certification Access	For NR & ReX staff	$9,000.00
Pilot Training Support	On-site sessions	$13,000.00
Staff Incentives & Engagement Materials	Communication campaigns	$5,000.00
Travel & Lodging (trainer travel)	Limited sessions	$5,000.00
Post-rollout Refreshers	3 months after go-live	$5,000.00
Testing & Quality Assurance		$35,000.00
Functional & Integration Testing	Build & pilot	$12,000.00
Data Validation & UAT	Pilot & rollout	$9,000.00
Security / Penetration Testing	Pre-rollout	$7,000.00
Performance / Load Testing	Full rollout	$7,000.00
Operations & Travel		$25,000.00
Project Meetings & Coordination	20 months	$8,000.00
Travel for On-site Work & Evaluation	6–8 trips	$8,000.00
Printing, Communication, Documentation	Materials & logistics	$4,000.00
Stakeholder Workshops & Review Meetings	3–4 sessions	$5,000.00
Contingency Reserve (10%)		$85,000.00
Risk-related or unplanned costs (vendor delays, rework, tool upgrades, etc.)	Lump sum reserve	$85,000.00

10.2 Payment and Funding Schedule



Milestone	Deliverable / Funding Trigger	Budget Release (%)
M1 – Project Initiation Approved	Charter sign-off and team formation	10 %
M3 – Design Approved	Design and planning documents finalised	15 %
M4 – Build and Migration Complete	Verified pilot data migration & testing	25 %
M5 – Pilot Evaluation Complete	Training delivered & pilot validated	25 %
M7 – Project Closure Approved	Final handover and sign-off	25 %

Payments are released internally by National Railways (NR) in alignment with milestone achievements confirmed by the Business Sponsor.

10.3 Cost Monitoring and Control

1. Cost Baseline: Defined in Microsoft Project; updated monthly against actual expenditure.
2. Variance Tracking: Monthly variance reports highlight deviations beyond ±10 %.
3. Change Control: Any scope-driven budget change requires sponsor approval.
4. Contingency Use: The 10 % reserve may be accessed only with written sponsor authorization.
5. Financial Reporting: Monthly internal reports and quarterly Steering Committee reviews.
6. Audit Trail: All financial records retained on NR's SharePoint for audit and accountability.

10.4 Financial Governance

Role	Responsibility
Project Manager (Kavish Gunasekara)	Tracks expenditure, prepares variance reports, and submits financial summaries.
Business Sponsor (NR COO)	Approves funding releases and ensures alignment with corporate objectives.
Finance Officer (NR Finance Dept.)	Manages fund transfers and keeps financial records.
Steering Committee	Reviews financial performance and authorises budget adjustments.

10.5 Financial Success Criteria

- Project delivered within ± 10 % of the approved budget.
- Annual cost savings of ≈ AUD 850 000 achieved post-implementation.
- Accurate and timely financial reporting maintained throughout the project.
- Sponsor and Steering Committee satisfaction with financial management processes.

10.6 Return on Investment (ROI)

The Return on Investment (ROI) measures the project's financial efficiency and benefit relative to its total cost.

For the NR & ReX Network Integration Project, the target is to achieve an ROI of 100 %, meaning that the project should return an amount equal to its investment cost.

ROI Formula

$$\text{ROI} = \left(\frac{\text{Net Profit}}{\text{Total Cost}}\right) \times 100$$

For an ROI of 100%, this means:

$$100 = \left(\frac{\text{Net Profit}}{\text{Total Cost}}\right) \times 100$$

Calculation:

Item	**Value (AUD)**	**Remarks**
Total Cost	850 000	Investment over 20 months
Target Net Profit	850 000	Savings within 1 year
ROI	**100 %**	Equal cost to benefit ratio
Payback Period	12 months	Full return achieved within first year post go live

Interpretation:

This means the project aims to save or generate benefits equal to the total investment within the projected four-year post implementation period (e.g., annual logistics savings of approximately AUD 850 000 × 4 years = AUD 3.4 million).

This confirms that the project is financially viable and sustainable, providing full cost recovery and long-term operational gains.

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